

Extracting Recipes from Cooking Videos using V-JEPA2 Embeddings

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Introduction

- ▶ Goal: extract **recipe steps** from egocentric cooking videos.
- ▶ Challenge: cluttered kitchens, occlusions, and background actions.
- ▶ Backbone: frozen **V-JEPA2** video embeddings.
- ▶ Classifier: lightweight temporal model on top of embeddings.

Two-stage pipeline

1. **Relevance filter**: cooking vs. background.
2. **Action classifier**: verb, noun, and verb-noun pair.

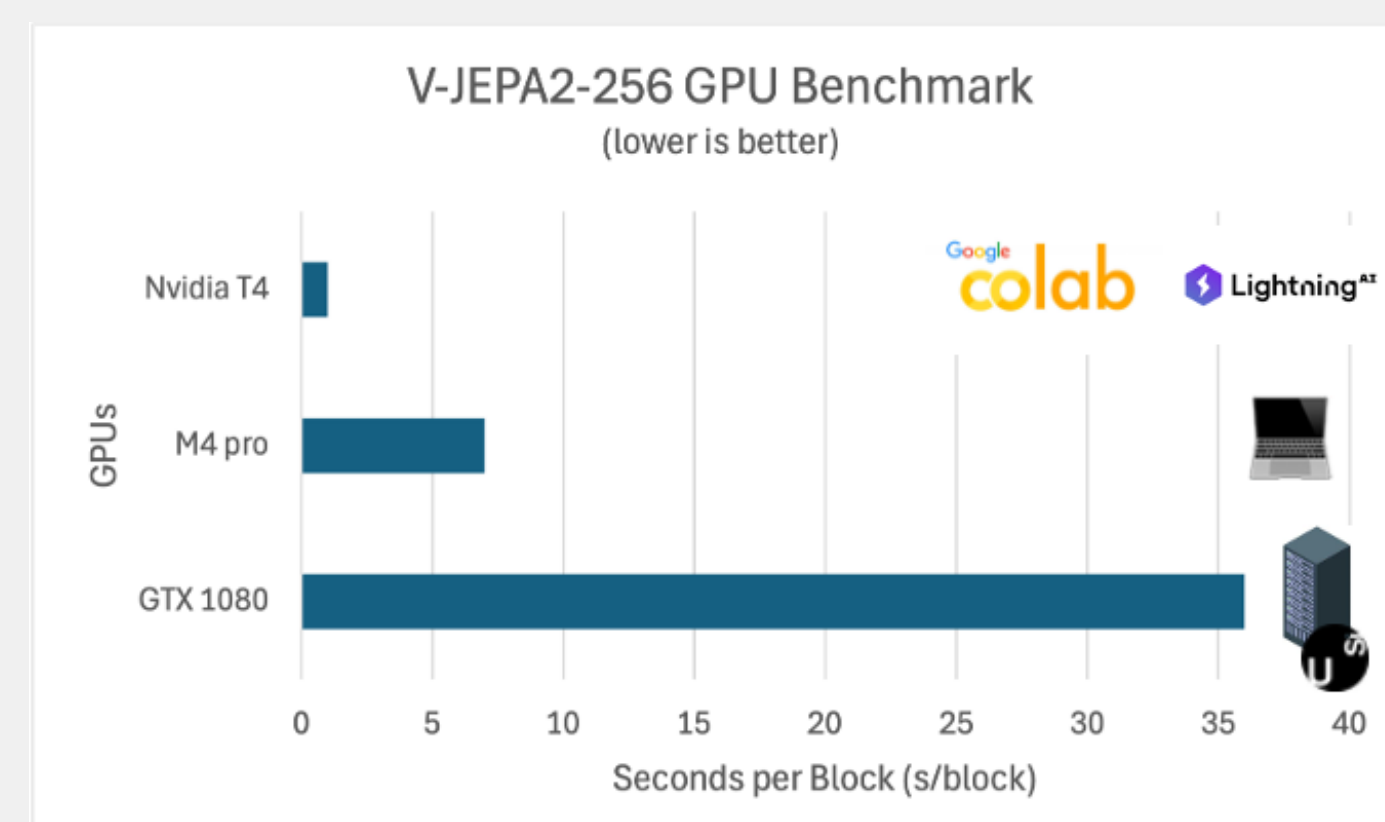
Dataset: EPIC-KITCHENS-100

- ▶ **408** embedded videos.
- ▶ **170,299** temporal blocks.
- ▶ **1024-D** V-JEPA2 embedding per 64-frame block.
- ▶ **116,895** blocks overlap an annotation.
- ▶ **53,404** background/unannotated blocks.
- ▶ **7,008** dataset action pairs; **2,721** action-head classes.

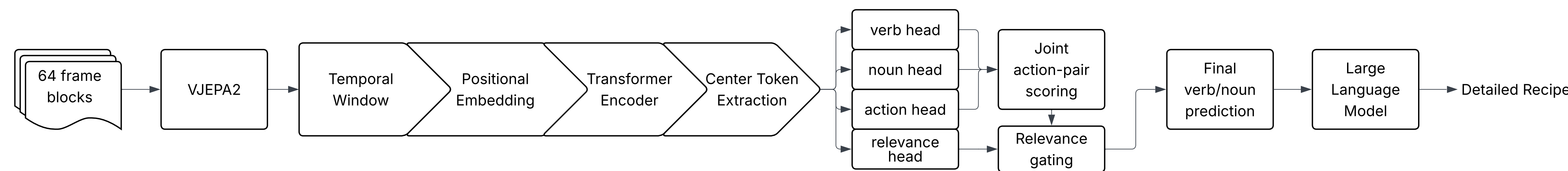
Block label	Fraction
Relevant	27.9%
Non-relevant	72.1%

Why V-JEPA2-small?

- ▶ **Small** V-JEPA2 variant.
- ▶ Frozen backbone: no video-model fine-tuning.
- ▶ Cloud compute for dataset embedding.
- ▶ Classifier training on Apple MPS.



Pipeline Overview



Classification Heads

Temporal encoder

- ▶ **7-block window** around target.
- ▶ Zero-padding at boundaries.
- ▶ 1024-D embeddings → **768-D**.
- ▶ Learned position embeddings.
- ▶ **2-layer, 8-head TransformerEncoder**.

Heads and decoding

- ▶ Center token feeds all heads.
- ▶ Heads: relevance, verb, noun, action pair.
- ▶ Relevance trains on **all** blocks.
- ▶ Action losses use annotated relevant blocks.
- ▶ Decoding adds pair prior, $w = 0.25$.
- ▶ Single-model and 3-model ensemble with averaged predictions.



Why a Transformer Encoder?

- ▶ Single blocks miss before/after context.
- ▶ Attention compares all 7 neighboring blocks.
- ▶ The model can see action setup and completion.
- ▶ Masking handles short windows at video boundaries.
- ▶ Only the center block is classified.

Results

Metric	Val	Test
Relevance accuracy	91.6%	92.3%
Relevance F1	84.8%	86.1%
Verb accuracy	94.4%	95.6%
Noun accuracy	98.3%	98.7%
Exact action	93.8%	95.3%

- ▶ Action rows: predicted relevant blocks for the ensemble.
- ▶ Oracle relevance: **91.8%** test exact action.
- ▶ Predicted relevant blocks: **95.3%** test exact action.

LLM Post-Processing

- ▶ Input: sequence of predicted **(verb, noun)** pairs.
- ▶ Model: **Llama 3.2**.
- ▶ Output: readable recipe steps.
- ▶ Repeated events are merged.
- ▶ Sparse predictions become natural instructions.

Conclusion

- ▶ Frozen V-JEPA2 embeddings reduce training cost.
- ▶ Relevance filtering removes background noise.
- ▶ Transformer context improves block-level decisions.
- ▶ Multi-head decoding predicts coherent verb-noun actions.
- ▶ LLM post-processing turns actions into recipes.